powerful new way of thinking about the theory. From the standpoint of numerical predictions, Feynman's perspective agrees exactly with all that went before. But its formulation is quite different. Let's describe it in the context of the electron two-slit experiment.

The troubling thing about Figure 4.8 is that we envision each electron as passing through either the left slit or the right slit and therefore we expect the union of Figures 4.4 and 4.5, as in Figure 4.6, to represent the resulting data accurately. An electron that passes through the right slit should not care that there also happens to be a left slit, and vice versa. But somehow it does. The interference pattern generated requires an overlapping and an intermingling between something sensitive to both slits, even if we fire electrons one by one. Schrödinger de Broglie, and Born explained this phenomenon by associating a probability wave to each electron. Like the water waves in Figure 4.7, the electron's probability wave "sees" both slits and is subject to the same kind of interference from intermingling. Places where the probability wave is augmented by the intermingling, like the places of significant jostling in Figure 4.7, are locations where the electron is likely to be found; places where the probability wave is diminished by the intermingling, like the places of minimal or no jostling in Figure 4.7, are locations where the electron is unlikely or never to be found. Electrons hit the phosphorescent screen one by one, distributed according to this probability profile, and thereby build up an interference pattern like that in Figure 4.8.

Feynman took a different tack. He challenged the basic classical assumption that each electron either goes through the left slit or the right slit. You might think this to be such a basic property of how things work that challenging it is fatuous. After all, can't you look in the region between the slits and the phosphorescent screen to determine through which slit each electron passes? You can. But now you have changed the experiment. To see the electron you must do something to it—for instance, you can shine light on it, that is, bounce photons off it. Now, on everyday scales photons act as negligible little probes that bounce off trees, paintings, and people with essentially no effect on the state of motion of these comparatively large material bodies. But electrons are little wisps of matter. Regardless of how gingerly you carry out your determination of the slit through which it